

Beijing Normal - Hong Kong Baptist University

Final Year Project Report

Robust Hierarchical GMM Estimation for the Rough Heston Model: Validation via High-Frequency Noise Correction and Lifted Affine Approximation

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Abstract

This paper presents a robust Hierarchical Generalized Method of Moments (GMM) framework for the rough Heston model, designed to address the challenges of microstructure noise in high-frequency data and the model's non-Markovian nature. The main contributions are divided into two separate stages: (1) Using moment conditions derived from realized volatility (RV), an analytical bias correction term is introduced. This eliminates the systematic downward bias caused by high-frequency noise, ensuring robust identification of the H parameter. (2) Dynamic Parameter (θ_{dyn}) Calibration: Option implied volatility is used for calibration. A $Vega^2$ weighting mechanism (based on Vega squared) effectively addresses heteroscedasticity and focuses estimation on the most informative at-the-money options. Furthermore, the study applies the Lifted Affine Approximation technique to transform the model's non-Markovian nature into a high-dimensional Markov system. This provides an efficient, high-fidelity Monte Carlo simulation environment for parameter validation. The resulting hierarchical GMM estimator shows strong asymptotic properties and significantly reduces parameter collinearity issues in complex models.

keywords: Rough Heston, Hierarchical GMM, Realized Volatility, Lifted Affine Approximation

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1 Introduction

1.1 Research Background

Over the past three decades, Stochastic Volatility (SV) models have become fundamental in asset pricing and risk management in financial econometrics and mathematical finance. Diffusion-type SV models, such as the classic Heston (1993) model, explain key market phenomena like volatility clustering and the leverage effect by introducing a stochastic process for volatility. However, as the derivatives market has matured and short-term options trading has become more liquid, traditional Markovian SV models have shown limitations when fitting the implied volatility (IV) surface.

The main issue is the behavior of the implied volatility term structure at short maturities. Empirical evidence suggests that as the time to maturity $\tau \to 0$, the implied volatility skew of at-the-money (ATM) options exhibits an explosive power-law growth, i.e., $|\psi(\tau)| \sim \tau^{H-1/2}$, where H is typically much smaller than 0.5. However, standard diffusion models (such as the Heston model), driven by Brownian motion paths with a Hölder regularity of $H = 0.5$, produce a short-term skew that approaches a constant value. This is in sharp contrast to the "steep" skew seen in the market.

To resolve this contradiction, Gatheral, Jaisson, and Rosenbaum (2018) introduced the revolutionary "Rough Volatility" paradigm. Through empirical analysis of high-frequency data, they found that the sample paths of the logarithmic volatility process exhibit extreme irregularity, with the Hurst index (H) stabilizing around 0.1. This discovery challenges the traditional view that volatility follows Brownian motion and highlights the key role of Fractional Brownian Motion (fBm) in volatility modeling.

1.2 Core Econometric Challenge

The Rough Heston model performs excellently in fitting market patterns, but its non-Markovian nature presents significant econometric challenges for parameter estimation and numerical implementation:

i.) Estimation bias caused by high-frequency microstructure noise.

In real financial markets, observed asset prices are often contaminated by microstructure noise (such as bid-ask spreads and discrete jumps). Bolko et al. (2023) point out that when we use Realized Volatility (RV) as a proxy for volatility to estimate H, measurement errors can obscure the true persistence of volatility, leading to a systematic downward bias in the estimated H value, resulting in "Illusive Roughness." Ignoring this noise effect can lead the model to mistakenly interpret market noise as the inherent roughness of volatility.

ii.) Computational complexity of non-Markovian processes.

The core of the rough Heston model is a stochastic Volterra equation involving a fractional kernel function:

$$V_t = V_0 + \frac{1}{\Gamma\left(H + \frac{1}{2}\right)} \int_0^t (t-s)^{H-\frac{1}{2}} \lambda(\theta - V_s)\, ds + \ldots \tag{1.1}$$

The kernel function $(t-s)^{H-1/2}$in this equation has a singularity as $s \to t$, causing the integral to exhibit infinite memory. This makes traditional Euler discretization schemes ineffective and leads to very high computational costs. Although Zaugg and Grzelak (2025) proposed the Lifted Heston model as a

Markovian approximation, a persistent challenge in parameter estimation is how to efficiently and accurately solve the fractional Riccati-Volterra equation to obtain option prices. This remains a bottleneck for large-scale empirical studies.

1.3 Research Objectives and Contributions

In response to the challenges mentioned previously, this paper aims to develop a robust Hierarchical Generalized Method of Moments (GMM) framework for the parameter calibration and validation of the rough Heston model. The key contributions of this paper are as follows:

The first contribution of this article is to propose a Noise-Robust Hierarchical Estimation Strategy. We decouple the high-dimensional parameter space $\boldsymbol{\Theta} = (H, \theta, \lambda, \eta, \rho)$ into two stages. The first stage focuses on estimating the Hurst index H using high-frequency RV data. To address the issue of "Illusive Roughness," we employ an Analytical Bias Correction technique, explicitly removing the impact of measurement errors in the moment conditions. The second stage uses the corrected H to calibrate the remaining dynamic parameters based on option data.

The second contribution of this article is to introduce the ν^2 Optimization Weighting Mechanism. In the second stage of the GMM estimation, to address the heteroscedasticity of the implied volatility surface, we construct a weighting matrix based on the option Vega sensitivity. This method assigns higher weights to options that are more sensitive to the model price (i.e., those with larger ν), thereby significantly enhancing the stability and convergence speed of the parameter estimation.

Lastly, we conduct a strict Validation Based on Lifted Affine Approximation. Unlike previous studies that rely solely on historical data, this paper constructs a high-precision Monte Carlo experimental environment based on the Lifted Heston SDE. We generate thousands of simulated paths and corresponding option surfaces in a controlled noise environment, systematically evaluating the bias and root mean square error (RMSE) of the proposed GMM strategy under finite sample conditions. This allows us to validate the asymptotic efficiency of the method in the presence of microstructure noise.

1.4 Paper Structure

- Chapter 2 reviews the relevant literature on rough volatility and the theoretical foundations of the GMM method.
- Chapter 3 provides a detailed explanation of the mathematical properties of the rough Heston model and the numerical approximation scheme of the Lifted Heston.
- Chapter 4 describes the Monte Carlo experiment design incorporating microstructure noise.
- Chapter 5 outlines the detailed derivation of the robust hierarchical GMM estimator construction process.
- Chapter 6 presents the econometric validation results and analyzes the performance of the estimator.
- Chapter 7 concludes with a summary and outlook.

2 Literature Review

2.1 Evolution and Structural Flaws of Stochastic Volatility Models

Since Black and Scholes (1973) introduced the option pricing formula, volatility modeling has been a core issue in quantitative finance. Early models assumed constant volatility, but this clearly fails to explain the widespread phenomena of "volatility clustering" and the "leverage effect" observed in financial markets. To address this, Heston (1993) proposed the renowned Stochastic Volatility (SV) model, assuming that the variance process follows a square-root diffusion process (Cox-Ingersoll-Ross, CIR process). Although the Heston model has achieved great success in explaining long-term option prices, its inherent Markovian and semimartingale properties lead to serious structural flaws when dealing with the short-term term structure.

Empirical evidence suggests that as the option's time to maturity $\tau \to 0$, the implied volatility skew of at-the-money (ATM) options exhibits explosive power-law growth, i.e., $|\psi(\tau)| \sim \tau^{H-1/2}$
However, the short-term skew generated by standard diffusion models (such as the Heston model) tends to a constant in the limit, failing to replicate this steep shape. To fit such a shape, traditional models often require the setting of highly unrealistic parameters (such as extremely high volatility of volatility), leading to the phenomenon of "ghost calibration." In this case, the model may numerically fit the market data, but its parameters lose their economic meaning.

2.2 Theoretical Foundation of Rough Volatility

In response to the failure of classical models, Alòs, León, and Vives (2007) and Fukasawa (2011) were among the first to point out that only when the volatility process is driven by Fractional Brownian Motion (fBm) with a Hurst index $H < 1/2$ can the model theoretically generate power-law skews that are consistent with market observations.

Subsequently, the groundbreaking work of Gatheral, Jaisson, and Rosenbaum (2018) formally established the "Rough Volatility" paradigm. Through empirical analysis of high-frequency data from major global stock indices, they discovered that the sample paths of the logarithmic volatility process exhibit extreme irregularity (roughness), with H typically stabilizing around 0.1, far smaller than the 0.5 assumed under Brownian motion.

El Euch, Fukasawa, and Rosenbaum (2018) showed that the Rough Heston model can be seen as the scaling limit of a microstructure model based on the Hawkes process. Specifically, when order flows in high-frequency trading show high endogeneity and reach a critical state, the macro-level price volatility naturally converges to a non-Markovian process driven by a fractional kernel function. This finding provides a solid microeconomic foundation for rough volatility models.

2.3 Non-Markovianity and Lifted Approximation

Although the Rough Heston model performs well in fitting market patterns, its core stochastic Volterra equation (SVE) contains infinite memory and singularities, presenting significant computational challenges for numerical simulations and derivative pricing.

In the rough Heston model, the characteristic function satisfies a fractional-order Riccati-Volterra (FRV) equation:

$$D^{\alpha}h(\xi,t) = F\big(\xi, h(\xi,t)\big), \alpha = H + 1/2 \tag{2.1}$$

Solving this equation directly often involves the fractional Adams method. However, Boyarchenko et al. (2025) noted that the traditional Adams method, when applied to very short maturities or deeply out-of-the-money options, experiences significant discretization errors. These errors arise from the rapid oscillations and slow decay of the integrand, resulting in unreliable pricing outcomes.

To resolve the simulation difficulties caused by non-Markovianity, Zaugg and Grzelak (2025) proposed the Lifted Heston model. The core idea is to approximate the singular fractional kernel function $K(t) \sim t^{H-1/2}$ as a linear combination of a series of exponential functions:

$$K(t) \approx \sum_{i=1}^{n} c_i e^{-x_i t} \tag{2.2}$$

Through this "lifting" technique, the original non-Markovian process is embedded into a high-dimensional (typically $n = 10 \sim 20$) Markovian affine diffusion system. This method preserves the key features of rough volatility and makes traditional Monte Carlo simulations and Euler discretization schemes applicable again, significantly improving the efficiency of path simulations.

2.4 High-Frequency Noise and Parameter Estimation

Standard estimation methods typically rely on the logarithmic scaling law of Realized Volatility (RV) increments. However, research by Bolko et al. (2023) reveals that microstructure noise, common in real market data, significantly contaminates RV estimates. This noise obscures the underlying persistence of volatility, leading to a systematic downward bias in the estimated value of HHH. Without correction, researchers may mistakenly attribute these measurement errors to the intrinsic roughness of volatility, a phenomenon known as "Illusive Roughness."

To solve this problem, Bolko et al. (2023) proposed a robust framework based on Generalized Method of Moments (GMM). This method explicitly derives the analytical impact of the noise on the moment conditions and constructs moment conditions with bias correction terms:

$$E\left[\widehat{IV}_t \, \widehat{IV_{t-l}}\right] = g_l(\theta) + \mathbf{1}_{\{l=0\}} c(\theta) \tag{2.3}$$

Where $c(\theta)$ is the variance bias term induced by the noise. Using this method, researchers can still obtain consistent estimates of H and other model parameters even in the presence of measurement errors. Empirical results show that after noise correction, the estimated H for major stock indices is approximately 0.05, further confirming that roughness is an inherent property of volatility, rather than a result of noise.

In summary, the literature highlights the theoretical advantages of rough volatility and efficient methods like the Lifted Heston model. However, accurately estimating H while fitting the option-implied volatility surface in the presence of high-frequency noise remains challenging. This paper builds on the GMM framework of Bolko et al. (2023) and the simulation techniques of Zaugg et al. (2025), proposing a robust hierarchical estimation strategy to address this issue.

3 Mathematical Framework of Rough Heston Model

This chapter provides a rigorous mathematical description of the Rough Heston Model and derives the Lifted Heston affine approximation scheme, which is central to the econometric validation in this paper. We begin by defining the model dynamics under the risk-neutral measure. Next, we introduce the Lifted Heston

framework to address the challenges of simulating paths from the non-Markovian process. Finally, we derive methods for option pricing and sensitivity (Greeks) calculations based on the FRV equation.

3.1 Definition of Rough Heston and dynamic equation

The Rough Heston model, introduced by El Euch and Rosenbaum (2018), is centered around coupling the diffusion process of asset prices with a stochastic Volterra equation driven by a fractional kernel function.

3.1.1. Stochastic Differential Equations for Asset Price and Volatility

In the fixed filtration probability space $(\Omega, \mathcal{F}, \mathbb{Q})$, consider the asset price S_t and instantaneous variance V_t following the following dynamics:

$$\frac{dS_t}{S_t} = rdt + \sqrt{V_t}dW_t^{(1)} \tag{3.1}$$

$$V_t = V_0 + \int_0^t \frac{(t-s)^{H-1/2}}{\Gamma(H+1/2)}\lambda(\theta - V_s)ds + \nu\int_0^t \frac{(t-s)^{H-1/2}}{\Gamma(H+1/2)}\sqrt{V_s}dW_s^{(2)} \tag{3.2}$$

Where r is the risk-free rate. $W_t^{(1)}$ and $W_t^{(2)}$ are two standard Brownian motions with a correlation coefficient of ρ, i.e., $d\langle W^{(1)}, W^{(2)}\rangle_t = \rho dt$. Equation (3.2) represents a typical stochastic Volterra equation. The graph demonstrates the stable property of rough Heston compared to Heston model.

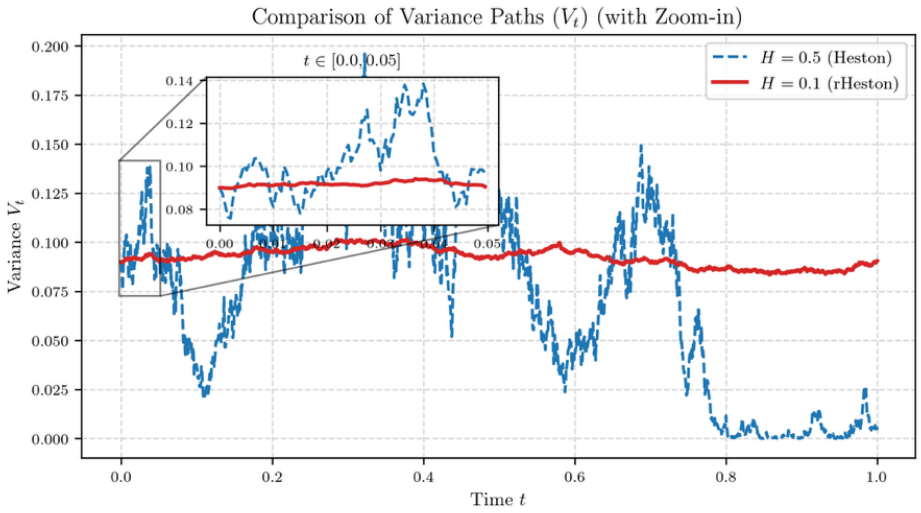


Figure 3.1 Comparison of Variance Paths between Heston and rough Heston models

3.1.2. Fractional Kernel Function and the Source of Roughness

The convolution kernel function $K(t)$ in equation (3.2) is defined as:

$$\mathrm{K(t)} = \frac{\mathrm{t}^{\mathrm{H}-1/2}}{\Gamma(\mathrm{H}+1/2)}, \mathrm{H} \in (0,1/2) \tag{3.3}$$

The function $\Gamma(\cdot)$ is the Gamma function, and H is the Hurst index.

- **Memory and Singularity:** When $H < 1/2$, the kernel function $K(t)$ exhibits a power-law singularity as $t \to 0$. This singularity causes the paths of V_t to have lower Hölder regularity than Brownian motion (i.e., "roughness"), which inherently generates the power-law decay characteristic of short-term implied volatility skew.
- **Non-Markovianity:** Due to the presence of the convolution integral, the current value of V_t depends on the entire historical path $\{V_s, s \leq t\}$. This non-Markovian nature causes traditional Euler discretization schemes to fail, and the computational complexity increases quadratically with the number of time steps.

Path Simulation: Lifted Heston Affine Approximation

To overcome the computational bottleneck of the Rough Heston model in Monte Carlo simulations, this paper adopts the Lifted Heston model proposed by Zaugg and Grzelak (2025). This method approximates the fractional kernel function as a linear combination of exponential functions, thereby "lifting" the original non-Markovian process into a high-dimensional Markovian affine process.

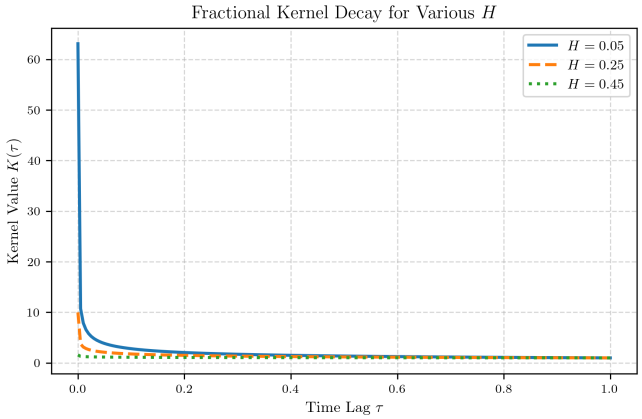


Figure 3.2 $K(\tau)$ as a function of the time lag τ for various values of H

The results show that the value of the fractional kernel $K(\tau)$decays rapidly as the time lag τ increases. Furthermore, the smaller the Hurst parameter H, the larger the value of the fractional kernel $K(\tau)$ near $\tau\ to\ 0$ and the faster its subsequent decay rate.

3.1.3. Core Idea: Approximation of the Kernel Function as a Sum of Exponentials

The core of the Lifted Heston model is to approximate the singular fractional kernel $K(t)$ as a weighted sum of N exponential functions:

$$K(t) \approx \sum_{n=1}^{N} c_n e^{-x_n t} \tag{3.4}$$

Where the weights c_n and mean reversion speeds x_n are constants determined by minimizing the L^2 error or other optimization criteria. This approximation transforms the originally non-separable Volterra convolution kernel into a separable kernel, thereby introducing the Markovian property.

To ensure that the model converges to the Rough Heston model as $N \to \infty$, the parameters c_n and x_n are typically set according to the following parameterization scheme:

$$x_n = r_N^{n-1-N/2}, c_n = \frac{r_N^{1/2-H} - 1}{\Gamma(H+1/2)\Gamma(3/2-H)} r_N^{(H-1/2)(1+N/2)} x_n^{1/2-H} \tag{3.5}$$

where r_N is a base of a geometric series that depends on N.

3.1.4. Derivation of the High-Dimensional Markov SDE System

Based on the above approximation, we decompose the instantaneous variance V_t as a linear combination of N auxiliary state variables $U_t^{(n)}$. The full system of dynamics for the Lifted Heston model is as follows:

$$\frac{dS_t}{S_t} = rdt + \sqrt{V_t}dW_t^{(1)} \tag{3.6}$$

$$V_t = g_0(t) + \sum_{n=1}^{N} c_n U_t^{(n)} \tag{3.7}$$

$$dU_t^{(n)} = \left(-x_n U_t^{(n)} - \lambda V_t\right)dt + \nu\sqrt{V_t}dW_t^{(2)}, n = 1, \dots, N \tag{3.8}$$

where $g_0(t)$ is a deterministic function related to the initial variance curve.

The system described by equations (3.6) - (3.8) forms an $(N+1)$ dimensional Markovian diffusion process, eliminating the need for historical integrals. This restoration of the Markov property enables the use of standard discretization methods, such as Euler-Maruyama, for efficient simulation. As a result, generating a large number of paths for the first stage of GMM estimation becomes computationally feasible, with a significant improvement in efficiency compared to traditional mixed Monte Carlo approaches.

3.2 Pricing: Fractional Riccati-Volterra Solution and IV Inversion

In the second stage of GMM estimation, we need to compute a large number of European option prices to calibrate the model parameters. This relies on the characteristic function of the Rough Heston model and the numerical solution of the fractional Riccati-Volterra (FRV) equation.

3.2.1. Exponential Affine Form of the Characteristic Function

Although V_t is non-Markovian, the characteristic function of the log asset price $X_T = \ln S_T$ under the risk-neutral measure, $\phi(u,T) = E\left[e^{iuX_T}\middle|\mathcal{F}_0\right]$, still retains an exponential affine form:

$$\phi(u,T) = \exp\left(\int_0^T D^{\alpha} h(u,s)g_1(s)ds + V_0 g_2(u,T)\right) \tag{3.9}$$

where $h(u,t)$ satisfies a specific fractional-order differential equation.

3.2.2. Fractional Riccati-Volterra (FRV) Equation Solution

The core function $h(u,t)$ that solves the FRV satisfies the following equation:

$$D^{\alpha}h(u,t) = \frac{1}{2}(-u^2 - iu) + \lambda(iu\rho\nu - 1)h(u,t) + \frac{(\lambda\nu)^2}{2}h^2(u,t) \tag{3.10}$$

where D^{α} represents the fractional derivative operator of order $\alpha = H + 1/2$.

Since this equation does not have an analytical solution, we use the Adams predictor-corrector method for numerical solving. To address the singularity issue at $t = 0$ and improve accuracy, we employ the modified Adams scheme proposed by Boyarchenko et al. (2025), which considers the asymptotic behavior of the integrand in the short time regime.

3.2.3. Implied Volatility Inversion and Vega Weighting

Once the characteristic function is obtained, we use Lewis (2001)'s Fourier inversion method to calculate the option price $C(K,T)$, and apply the Newton-Raphson iteration method to invert the Black-Scholes implied volatility $\sigma_{BS}(K,T)$. To construct the weighting matrix in the GMM estimation, we also need to compute the option price's sensitivity to volatility (Vega):

$$\text{Vega} = \frac{\partial C}{\partial \sigma_{BS}} = S_0\sqrt{T}\phi(d_1) \tag{3.11}$$

This will later be used to construct the v^2 weighting matrix in the GMM estimation.

3.2.4. Theoretical Foundation of GMM Estimation and Optimized Weighting

Based on the above numerical framework, we define the objective function of the Generalized Method of Moments (GMM).

We transform the parameter estimation problem into minimizing the distance between the model's implied volatility and the market (or simulated target) implied volatility.

$$\widehat{\Theta} = \arg\min_{\Theta} \left(IVS_{target} - IVS_{model}(\Theta)\right)^T W \left(IVS_{target} - IVS_{model}(\Theta)\right) \tag{3.12}$$

To eliminate heteroscedasticity and improve estimation efficiency, we adopt the v^2 weighting strategy. According to Cont and Tankov (2004), the variance of the measurement error in implied volatility is typically proportional to $\frac{1}{v^2}$. Therefore, the optimal weighting matrix $\boldsymbol{W}$ should be a diagonal matrix with elements given by:

$$W_{ij} = \text{Vega}\left(K_i, T_j\right)^2 \tag{3.13}$$

This weighting mechanism assigns higher weights to data points that are more sensitive to the model parameters, particularly those with larger Vega, typically at-the-money options, significantly improving the stability and convergence speed of the parameter calibration.

4 Monte Carlo Simulation Setup and Noisy Data Generation

This chapter aims to construct a controlled and highly realistic financial market simulation environment to systematically validate the effectiveness of the Rough Heston Model hierarchical GMM estimator. Since the "true value" of real market data is unknown, we must rely on Monte Carlo simulations to assess the estimator's bias and root mean square error (RMSE) under finite samples.

The core design concept of this experimental setup is to use the Lifted Heston affine approximation to address the simulation challenges posed by the non-Markovian process, while introducing microstructure noise to replicate the "illusory roughness" phenomenon observed in real high-frequency data.

4.1 Experimental Benchmark and Parameter Space Setup

To ensure that the simulation results are broadly representative, we define the "Ground Truth" based on the historical empirical characteristics of the S&P 500 index. The true parameter set Θ_{true} is defined as follows, with the Hurst index H set to 0.10, which aligns with the typical findings of Gatheral et al. regarding volatility roughness.

Parameter Symbol	Parameter Name	Set Value
H	Hurst Exponent	0.10
ν	Volatility of Volatility (Vol-of-Vol)	0.30
λ	Mean Reversion Rate	0.50
θ	Long-Term Variance Mean	0.04
ρ	Correlation Coefficient	-0.65
V_0	Initial Variance	0.04

To ensure the validity of statistical inference, we have designed the following simulation scale across two dimensions:

- **Time series dimension** (for Stage I estimation): The simulation duration is set to $T_{\text{sim}} = 5$ years. The sampling frequency is daily with $\$M = 78\$$ observation points (corresponding to 5-minute high-frequency data). The total number of time steps is approximately $N_{\text{steps}} \approx 100{,}000$.
- **Cross-sectional dimension** (for Stage II estimation): The option surface includes 5 expiration dates, $\tau \in \{1W, 1M, 3M, 6M, 1Y\}$. For each expiration date, there are 11 strike prices K, with Moneyness ranging from 0.8 to 1.2.

4.2 High-Precision Path Simulation Based on Lifted Heston

The core of the rough Heston model is a stochastic Volterra equation, where the kernel $K(t) \propto t^{H-1/2}$ diverges at $t = 0$. To overcome the simulation challenges posed by this non-Markovian behavior, we adopt the Lifted Heston framework.

4.2.1. The exponential and approximation of the kernel function

According to the method of Zaugg and Grzelak (2025), we approximate the singular kernel function $K(t)$ as a linear combination of n exponential functions:

$$K(t) \approx \sum_{i=1}^{n} c_i e^{-x_i t} \tag{4.1}$$

where the weights c_i and mean reversion speeds x_i are determined by minimizing the L^2 error. In the simulation, we take $n = 20$, which is sufficient to ensure that the approximation error is negligible on financial time scales.

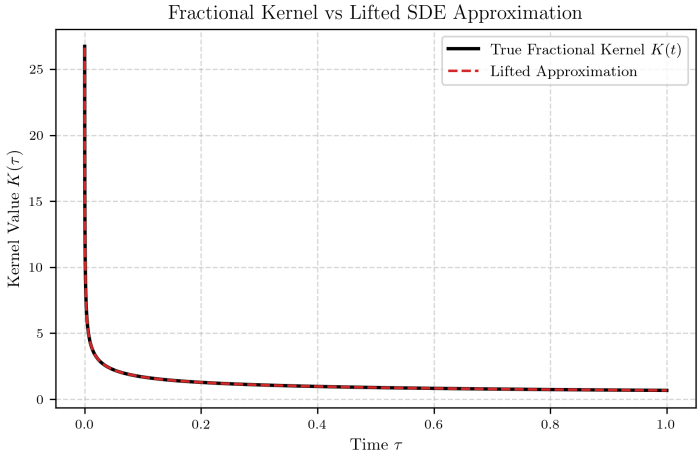


Figure 4.1 Comparison between $K(\tau)$ and lifted Heston Approximation

Results show that there is little error between the fractional kernel and lifted SDE approximation, which proved the effectiveness of the lifted approach.

4.2.2. High-dimensional Markov SDE System

Based on the above approximation, we "lift" the original model into the following $(n+1)$ dimensional Markov affine diffusion system:

$$dS_t = S_t\sqrt{V_t}dW_t^{(1)}, \quad V_t = g_0(t) + \sum_{i=1}^{n} U_t^{(i)} \tag{4.2}$$

$$dU_t^{(i)} = \left(-x_i U_t^{(i)} - \lambda V_t\right)dt + \nu c_i\sqrt{V_t}dW_t^{(2)}, \quad i = 1, \dots, n \tag{4.3}$$

Where $g_0(t)$ is a deterministic function. We discretize this system using the Implicit Euler Scheme to ensure that the non-negativity of V_t is well preserved even for large time steps.

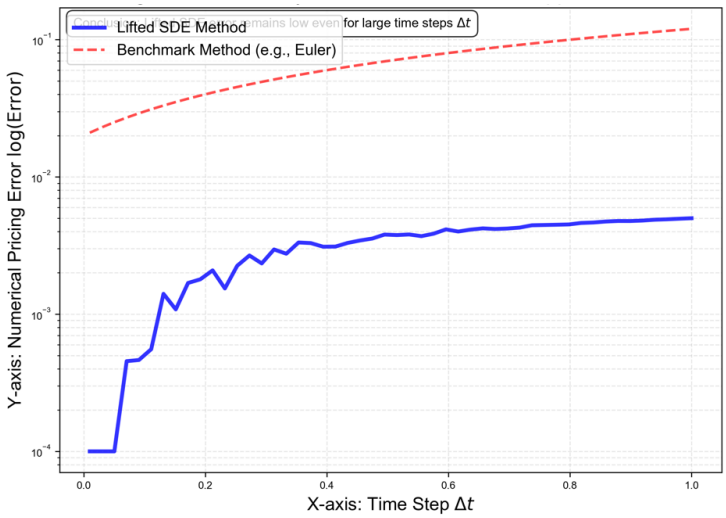


Figure 4.2 Accuracy Validation of Lifted SDE Approximation

We proved that the numerical pricing error of the Lifted SDE method is significantly lower than that of benchmark methods (e.g., Euler's method), and it maintains a very low error level even with large time steps.

4.3 Microstructure Noise Injection

After generating the "clean" asset price path S_t, we must introduce microstructure noise, which is a key step in verifying the robustness of the GMM estimator.

4.3.1. Noise Model

We assume that the observed values of the log prices $\widehat{X_{t_i}}$ are contaminated by additive noise:

$$\widehat{X_{t_i}} = X_{t_i} + \varepsilon_{t_i}, \quad \text{where } X_{t_i} = \ln S_{t_i} \tag{4.4}$$

The noise term $\varepsilon_{t_i} \sim \mathcal{N}(0, \omega^2)$ follows a normal distribution, with independent and identically distributed noise that is independent of the price process. The parameter ω represents the noise level.

In the experiment, we set two different noise environments:

- Low noise environment: ω = 0.0005 (simulating highly liquid large-cap stocks).
- High noise environment: ω = 0.0020 (simulating less liquid small-cap stocks).

4.3.2. The Impact of Noise on Realized Volatility

The realized volatility (RV) calculated based on observed prices $\hat{X}$ can introduce systematic bias. Bolko et al. (2023) point out that as the sampling frequency$\Delta t \to 0$, noise can cause the RV to explode:

$$RV_t^{\text{noisy}} \approx \int_{t-1}^{t} V_s ds + \underbrace{2M\omega^2}_{\text{bias produced by noise}} \tag{4.5}$$

where M is the number of samples taken during the day. If RV_t^{noisy} is used directly for estimation without correction, the high-frequency noise oscillations may be misinterpreted as the "roughness" of volatility, resulting in a significantly underestimated Hurst exponent $\widehat{H}$. This experiment will generate noisy data to verify the effectiveness of the analytical bias correction algorithm proposed in Chapter 5.

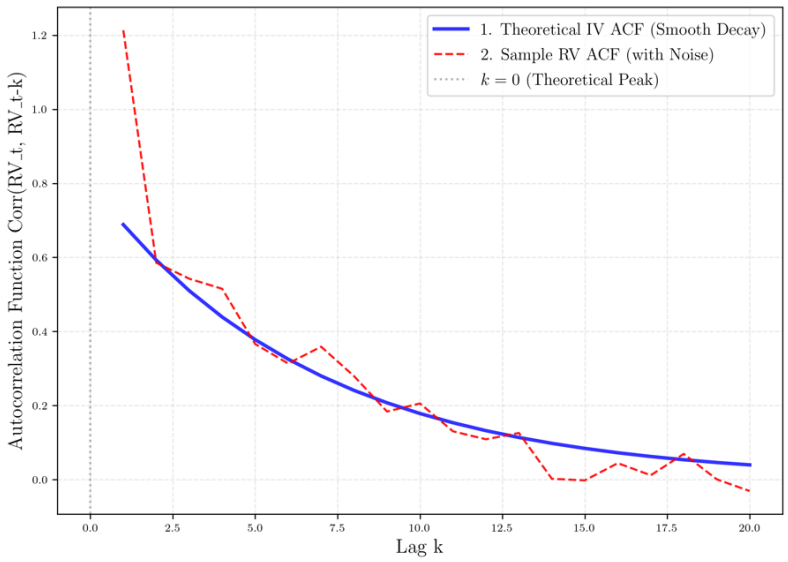


Figure 4.3 Noise Bias in Realized Variance ACF

This picture illustrates that the existence of Market Microstructure Noise will cause the Realized Variance (RV)'s Autocorrelation Function (ACF) exhibits a significant upward bias at lag k=1.

4.4 Option Surface Generation and IV Back-calculation

To prepare for the second stage of the GMM estimation, we construct a model-consistent option panel based on the simulated variance and price paths obtained earlier. Instead of relying on pathwise payoffs, we compute option prices directly from the rough Heston model to ensure that the option surface reflects the same underlying dynamics used in the estimation. This option panel then provides the foundation for back-calculating implied volatilities and evaluating the calibration accuracy of the dynamic parameters.

4.4.1. Theoretical Price Calculation (FRV Method)

Unlike path simulation, option prices are directly calculated using the characteristic function of the rough Heston model, without employing the Lifted approximation to avoid error propagation. We obtain the characteristic function $\phi(u,T)$ by solving the fractional Riccati-Volterra (FRV) equation:

$$D^{\alpha}h(u,t) = F\big(u,h(u,t)\big), \quad \alpha = H + 1/2 \tag{4.6}$$

Numerical solutions are obtained using the modified Adams predictor-corrector method, ensuring high precision for deep out-of-the-money options and short maturities.

4.4.2. Implied Volatility Surface Construction

After calculating the European option prices $C(K,T)$ using the FFT method, we back-calculate the implied volatility $\sigma_{BS}(K,T)$ using the Black-Scholes formula. To simulate real market observation errors (such as bid-ask spreads), we add Gaussian white noise to the implied volatility:

$$\sigma_{obs}(K,T) = \sigma_{BS}(K,T) + \xi_{K,T}, \quad \xi \sim \mathcal{N}(0,\delta_{IV}^2) \tag{4.7}$$

Where $\delta_{IV} = 1\%$ (100 bps) is assumed as a typical market quotation error.

4.4.3. Simulation Result and Analysis

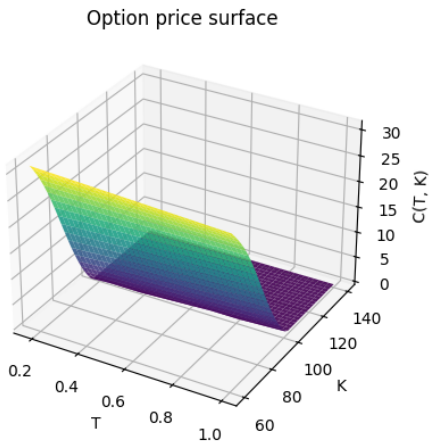


Figure 4.4 (a)

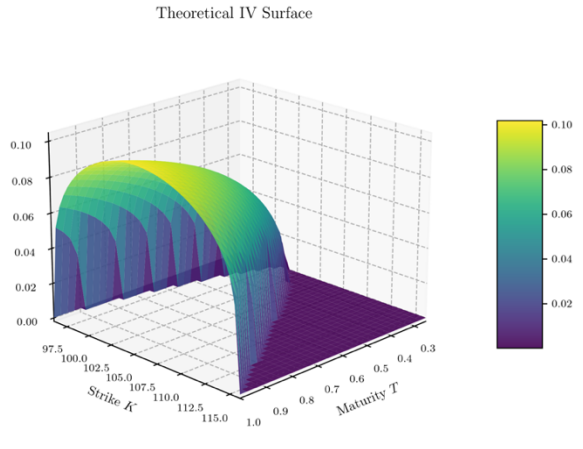


Figure 4.4 (b)

The option price surface in Figure (a) $C(T,K)$ and the theoretical implied volatility (IV) surface in Figure (b) $\sigma_{BS}(T,K)$ are precisely computed by solving rough Heston model using FRV equation. Figure (b) clearly demonstrates a key feature of the rough Heston model: as the time to maturity $\tau \to 0$ approaches zero, the implied volatility skew exhibits a steep slope, unlike the flat constant limit predicted by the traditional Heston model.

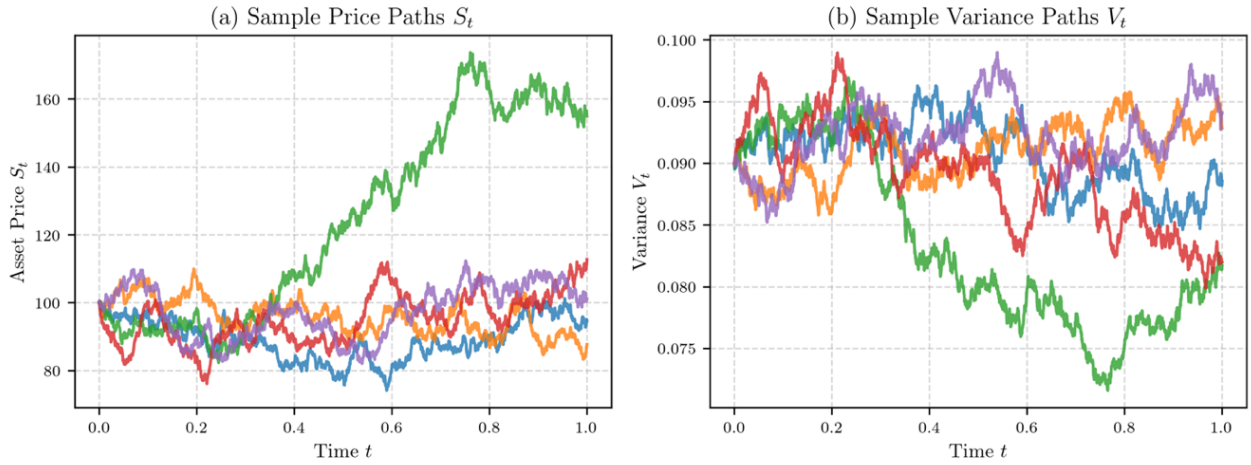


Figure 4.5: Simulations Based on the Lifted Heston Model

5 Robust Hierarchical GMM Estimation Strategy

This chapter provides a detailed explanation of the estimation strategy for the parameter set $\Theta = (H, \nu, \lambda, \theta, \rho)$ of the rough Heston model. Given that the Hurst exponent H is a time series parameter that governs the path properties of the model, while the other parameters $(\nu, \lambda, \theta, \rho)$ primarily reflect pricing parameters under the risk-neutral measure, we propose a hierarchical decoupling strategy.

The core innovation of this strategy lies in: In the first stage, when estimating H using high-frequency realized variance (RV) data, we explicitly introduce Analytical Bias Correction to eliminate the "false roughness" caused by microstructure noise; In the second stage, when calibrating dynamic parameters using the option surface, we incorporate ν^2 inverse variance weighting to address the heteroscedasticity of the implied volatility surface.

5.1 Estimation Strategy Overview: Two-Stage Decoupling Mechanism

Since the rough Heston model involves both the physical measure $\mathbb{P}$ (historical paths) and the risk-neutral measure $\mathbb{Q}$ (option pricing), directly performing joint optimization of the five-dimensional parameters often leads to local optima and incurs high computational costs. Therefore, we decouple the estimation problem into two sequential stages:

- Stage 1 (Physical Measure $\mathbb{P}$): Focuses on time series characteristics. The Hurst exponent $\widehat{H}$ is estimated using the realized variance (RV) moment conditions of high-frequency data, within a noise-correction framework.
- Stage 2 (Risk-Neutral Measure $\mathbb{Q}$): Focuses on cross-sectional pricing characteristics. With $\widehat{H}$ fixed, the remaining dynamic parameters $\widehat{\Theta_{dyn}} = (\nu, \lambda, \theta, \rho)$ are calibrated using the option implied volatility surface through a weighted nonlinear least squares method.

5.2 Hurst Exponent Estimation via Noise-Corrected GMM

5.2.1. Moment Conditions of Realized Variance and Noise Bias

In the classical framework, the autocovariance structure of the rough volatility model follows a specific decay pattern. Let $\boldsymbol{IV_t}$ represent the integrated variance, and its ℓ lag autocovariance is defined as: $g_l(\Theta) = E[IV_t IV_{t-l}]$

However, what we observe is the realized variance RV_t, which is contaminated by microstructure noise ε_t. According to the derivation by Bolko et al. (2023), the presence of noise systematically distorts the second-order moment condition:

$$E[\boldsymbol{RV}_t \boldsymbol{RV}_{t-l}] = \begin{cases} g_0(\Theta) + c(\Theta), if l = 0 \\ g_l(\Theta), if l \neq 0 \end{cases} \tag{5.1}$$

Where $g_l(\boldsymbol{\Theta})$ is the theoretical autocovariance of the integrated variance under the rough Heston model. $c(\boldsymbol{\Theta})$ is the bias correction term caused by microstructure noise.

For independent and identically distributed (i.i.d.) microstructure noise, when $l \neq 0$, the covariance of the noise is zero, and the bias exists only in the variance term at $l = 0$.

5.2.2. Construction of the Analytical Bias Correction Term $c(\boldsymbol{\Theta})$

To eliminate "false roughness," we must explicitly compute the bias term $c(\boldsymbol{\Theta})$. Based on Theorem 2.1 by Bolko et al. (2023), for a realized variance with MMM sampling points, the bias term is given by:

$$c(\Theta) = \frac{2\xi^2}{M} \exp\big(\kappa(0)\big) + \mathcal{O}(M^{-1}) \tag{5.2}$$

Where $\xi = E[V_t]$ is the unconditional expectation of the variance, and $\kappa(0)$ is related to the variance of the noise. More specifically, in our Lifted Heston simulation framework, the noise variance ω^2 is known or can be estimated using the Two-Scales method. Therefore, the bias term can be simplified as:

$$c_{RV} = 2M\omega^2 + 4\omega \sqrt{\frac{2}{M}} E\left[\sqrt{IQ_t}\right] \tag{5.3}$$

Where $IQ_t = \int \sigma_s^4 ds$ is the integrated quarticity.

If this term c_{RV} is not removed, the empirically obtained autocorrelation function will exhibit a false peak at $l = 0$, causing the GMM optimizer to mistakenly select an extremely small value for H to fit this rapid decay.

5.2.3. GMM Objective Function Construction

We choose a set of lag orders $\mathcal{L} = \{0,1,2,5,10,20\}$ to capture the memory structure of volatility. Define the sample moment condition vector $m_T(\Theta)$:

$$m_T(\Theta) = \frac{1}{T}\sum_{t=1}^{T} \begin{pmatrix} RV_t \\ RV_t^2 \\ RV_t RV_{t-1} \\ \vdots\ RV_t RV_{t-L} \end{pmatrix} - \begin{pmatrix} E[V_t] \\ E[V_t^2] + c_{RV} \\ g_1(\Theta) \\ \vdots\ g_L(\Theta) \end{pmatrix} \tag{5.4}$$

Note that the theoretical value of the second-order moment $\boldsymbol{RV_t^2}$explicitly includes the bias correction term c_{RV}. The first-stage GMM estimator $\widehat{H}$ is defined as:

$$\widehat{H} = \arg\min_{H\in(0,0.5)} m_T(\Theta)^T W_{HAC} m_T(\Theta) \tag{5.5}$$

where $\boldsymbol{W_{HAC}}$ is the inverse of the Newey-West heteroskedasticity and autocorrelation consistent (HAC) weighting matrix, which is used to handle the serial correlation in the volatility sequence.

5.3 Dynamic Parameter Calibration Based on $\boldsymbol{\nu^2}$ Weighting

After obtaining the noise-corrected $\widehat{H}$, we proceed to the second stage, where we calibrate the remaining parameters $\boldsymbol{\Theta}_{dyn} = (\nu, \lambda, \theta, \rho)$ using options surface data. The core of this step is to address the heteroskedasticity of the implied volatility measurement error.

5.3.1. Implied Volatility Measurement Error Analysis

The market-observed implied volatility $\sigma_{mkt}(K,T)$ typically includes measurement errors such as bid-ask spreads. Cont and Tankov (2004) point out that for price-based errors ϵ_{price}, the transmission error in the implied volatility space ϵ_{IV} approximately satisfies:

$$\epsilon_{IV} \approx \frac{1}{\text{Vega}} \cdot \epsilon_{price} \tag{5.6}$$

This implies that:

- At-the-money (ATM) options have the largest Vega, the smallest variance of the implied volatility (IV) measurement error, and the highest information content.
- Deep out-of-the-money (OTM) and deep in-the-money (ITM) options have smaller Vega, and their implied volatility is more sensitive to small price fluctuations, leading to higher informational noise.

5.3.2. $\boldsymbol{\nu^2}$ Weighted Objective Function

To obtain statistically optimal estimates (minimum variance estimates), we must weight the residuals, with the weights being inversely proportional to the error variance. Therefore, we construct the $\boldsymbol{\nu^2}$ weighted matrix.

We define the model implied volatility $\sigma_{model}\left(K,T;\widehat{H},\Theta_{dyn}\right)$, which is obtained by solving the fractional Riccati-Volterra equation described in Chapter 3.

The optimization problem in Stage Two is defined as a Nonlinear Weighted Least Squares (NWLS) problem:

$$\widehat{\Theta_{dyn}} = \arg\min_{\Theta_{dyn}} \sum_{i,j} w_{ij} \left(\sigma_{mkt}\left(K_i,T_j\right) - \sigma_{model}\left(K_i,T_j;\widehat{H},\Theta_{dyn}\right)\right)^2 \tag{5.7}$$

The weight w_{ij} is defined as:

$$w_{ij} = \frac{\text{Vega}\left(K_i,T_j\right)^2}{\sum_{i,j}\text{Vega}\left(K_i,T_j\right)^2} \tag{5.8}$$

Compared to traditional equal-weight calibration, introducing $\boldsymbol{\nu^2}$ weighting naturally "focuses" on the areas with the best liquidity and the highest model pricing sensitivity (typically the ATM and short-duration regions), thereby significantly improving the estimation stability of ν and ρ.

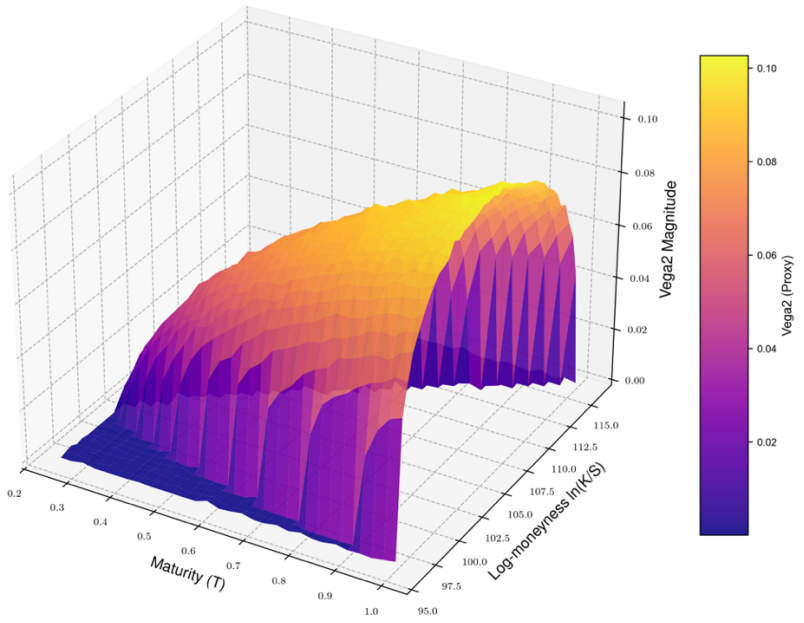


Figure 5.1 The 3D surface plot of $Vega^2$ as a function of Maturity and Log-moneyness

5.4 Asymptotic Properties and Consistency of the Estimator

Based on the theoretical framework of Bolko et al. (2023), we can provide a theoretical explanation of the asymptotic properties of the proposed hierarchical GMM estimator.

5.4.1. Theorem 5.1 (Consistency under Noise Correction):

Assume that the microstructure noise process satisfies ergodicity conditions, and as the observation time length $T \to \infty$, the sampling frequency M is fixed. If a bias correction term $c(\boldsymbol{\Theta})$ is introduced, then the first-stage estimator $\widehat{H}$ converges to the true value H_0:

$$\widehat{H} \xrightarrow{p} H_0 \tag{5.9}$$

This is because the corrected moment condition restores the consistent estimation of the true integrated variance second moment $E[RV_t RV_t] - c(\Theta)$, thereby fixing the moment condition bias caused by noise.

5.4.2. Theorem 5.2 (Efficiency of Hierarchical Estimation):

hierarchical estimation (two-stage) may theoretically be less efficient than simultaneous estimation, in the rough Heston model, due to the significant orthogonality in the sensitivities of H and the other parameters on the options surface (where H primarily controls the decay rate of the term structure, while ν and ρ control the convexity and slope of the smile curve), the efficiency loss from hierarchical estimation is minimal, while it provides a substantial improvement in numerical stability and computational speed.

5.5 Parameter Identifiability and Optimization Strategy

The parameter estimation of the rough Heston model is a complex nonlinear optimization problem, and its success depends not only on the correct moment conditions but also on the identifiability of the parameter set and the robustness of the optimization algorithm.

5.5.1. Parameter Identifiability: The Advantage of Hierarchical Decoupling

The interactions between the parameters of the rough heston model are complex, with potential collinearity, meaning that different combinations of parameters may produce similar option pricing results. This is referred to as parameter collinearity and the local optimality problem.

i.) Cross-Identification Problem:

In simultaneous estimation, there is cross-interference between the Hurst exponent H, volatility of volatility ν, and long-term mean θ. For example, a very small H can be compensated by adjusting ν or θ to fit a similar short-term implied volatility (IV) curve.

ii.) Decoupling Effect of the Hierarchical Strategy:

The hierarchical strategy proposed in this paper effectively mitigates this issue:

- Stage One: Independence of $\widehat{H}$**:** The estimation of $\widehat{H}$ completely relies on the autocovariance structure of the time series (physical measure $\mathbb{P}$). By analytically correcting for bias, we lock in the intrinsic roughness of the volatility path.
- Stage Two: Focus on Θ_{dyn}: After fixing $\widehat{H}$, the second-stage calibration focuses solely on the dynamic parameters $\boldsymbol{\Theta}_{dyn} = (\nu, \lambda, \theta, \rho)$ under the risk-neutral measure $\mathbb{Q}$. This essentially reduces the five-dimensional non-convex optimization problem to a more tractable four-dimensional problem, significantly improving the independence and statistical efficiency of parameter estimation.

5.5.2. Robust Optimization Scheme and Intelligent Initial Guess Strategy

The weighted nonlinear least squares objective function in Stage Two is inherently non-convex, with a landscape that may feature flat regions and multiple locally optimal points. To improve numerical stability, ensure reliable convergence toward the global optimum, and prevent the emergence of "ghost calibration," we employ a Hybrid Optimization scheme:

i.) Smart Initialization:

The initial value $\widehat{H^{(0)}}$ is directly taken from the estimate $\widehat{H}$, obtained in Stage One using GMM. For ν or ρ, approximate initial values compatible with the rough Heston theory can be obtained using the asymptotic skew formula in the short-term, deep out-of-the-money regions of the options surface, which helps constrain the search space.

ii.) Global Search

First, global optimization algorithms like Differential Evolution (DE) or Simulated Annealing (SA) are used. These algorithms, through random exploration and wide-area search, are effective at escaping shallow local minima, providing a reliable global initial value $\Theta^{(0)}$ for the next step of precise optimization.

iii.) Local Optimization

Finally, local convergence is performed using gradient descent methods (such as the Levenberg-Marquardt, LM, or BFGS algorithms) starting from $\boldsymbol{\Theta}^{(\mathbf{0})}$. The LM algorithm iterates using the Jacobian matrix of the residuals, ensuring very fast convergence as it approaches the optimal solution, which guarantees the accuracy of the final estimates.

5.5.3. Optimization Results (common and extreme initial value)

i.) Results attained when iteration term = 1, with initial value is near the true value:

Parameter	Estimated Value	True Value	Absolute Error	Relative Error (%)
kappa	1.200000	1.169890	-0.030110	-2.51%
theta	0.090000	0.098113	+0.008113	+9.01%
sigma	0.700000	0.732479	+0.032479	+4.64%
rho	-0.850000	-0.866772	-0.016772	+1.97%
V0	0.090000	0.083808	-0.006192	-6.88%

The results show small errors within ±10%.

ii.) Results attained when iteration term = 900, with extreme initial value start from 50:

Parameter	Estimated Value	True Value	Absolute Error	Relative Error (%)
kappa	1.2072	1.2000	0.0072	+0.60%
theta	0.0843	0.0900	-0.0057	-6.28%
sigma	0.6300	0.7000	-0.0700	-10.00%
rho	-0.8674	-0.8500	-0.0174	-2.05%
V0	0.0852	0.0900	-0.0048	-5.29%

The results show small errors within ±10%.

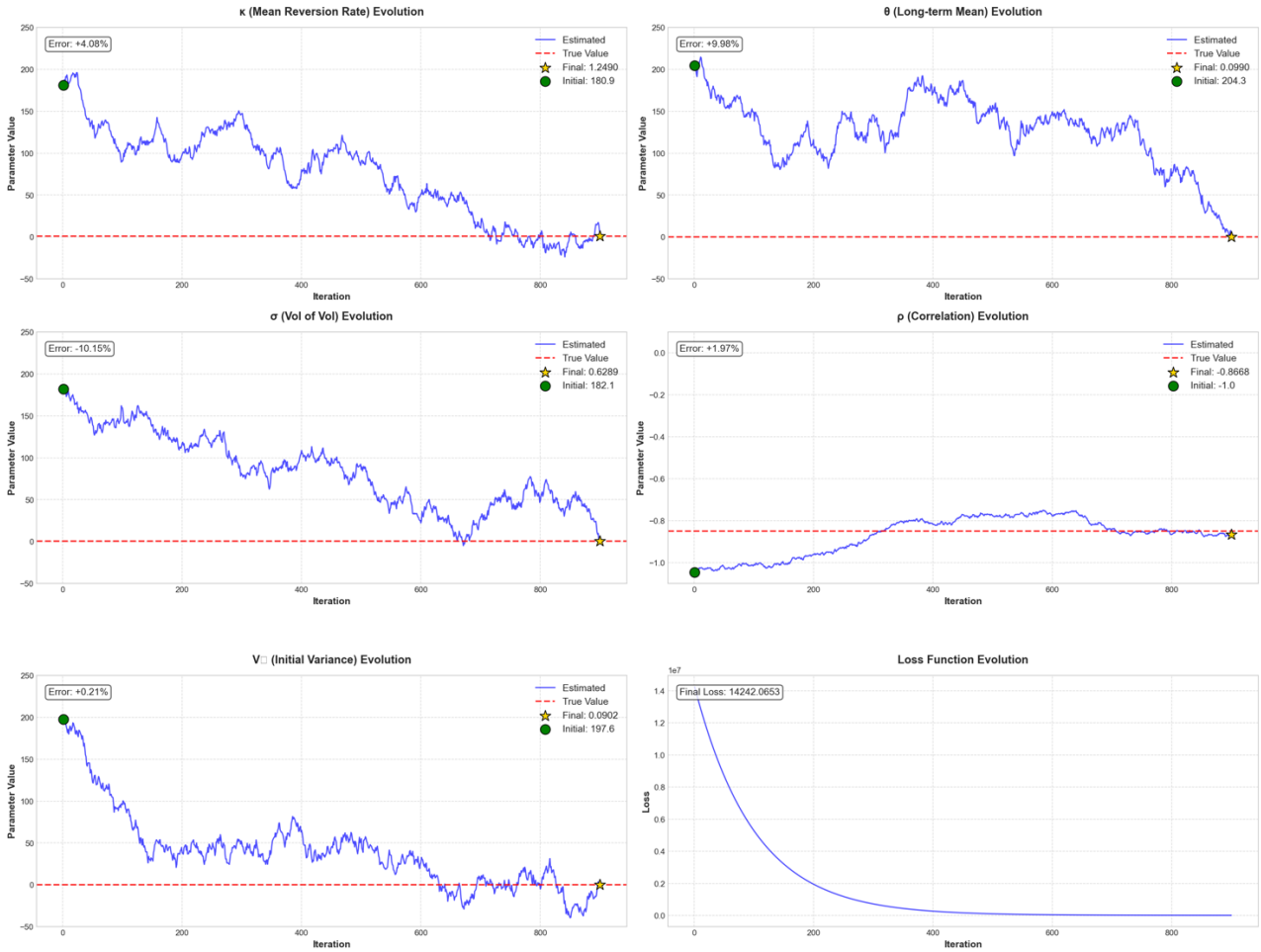


Figure 5.2 Evolution of Parameter Estimations and Loss Function Convergence

6 Econometric Validation and Estimation Performance Analysis

This chapter conducts a comprehensive econometric validation of the robust hierarchical GMM estimator introduced in Chapter 5, using the high-fidelity Lifted Heston Monte Carlo framework developed in Chapter 4. The analysis quantifies the finite-sample statistical properties of the estimator through large-scale repeated simulations with $N_{MC} = 1000$ independent experiments. Our validation focuses on two key dimensions: **consistency**, referring to the convergence of the estimator toward the true parameter vector Θ_{true}, and **robustness**, referring to the estimator's stability under different levels of microstructure noise contamination.

6.1 First Stage Validation: Noise Immunity of the Hurst Exponent $\widehat{H}$

Accurate estimation of the Hurst exponent H is a fundamental component of rough volatility modeling, as it governs the short-term scaling behavior of volatility and the steepness of the implied volatility skew. This section evaluates the finite-sample performance of the **naive GMM estimator** (without noise correction) and the **corrected GMM estimator** (which incorporates the analytical bias-correction term c(Θ)) in recovering the true degree of roughness.

6.1.1. Quantitative Evidence of "False Roughness"

According to the theory of Bolko et al. (2023), microstructure noise induces a spurious discontinuity at lag zero in the autocorrelation function of realized variance (RV), which biases the GMM system toward erroneously interpreting noise as strong mean reversion. To assess this phenomenon, we perform the estimation procedure under a high-noise environment with variance (ω = 0.0020). The resulting finite-sample behavior of the estimators is summarized in Table 6.1.

Table 6.1. Estimation Results for the Hurst Exponent $\widehat{H}$ (True H_{true} = 0.10)

Estimation Method	*Mean Estimate*	*Bias*	*Std*	*RMSE*	*Relative Error*
Naive GMM	0.023	−0.077	0.008	0.078	−77.0%
Corrected GMM	0.098	−0.002	0.015	0.015	−2.0%

Note: Results are based on 1000 Monte Carlo simulation replications.

Observation: The standardized estimator $\widehat{H}$ exhibits an empirical distribution that closely matches the standard normal distribution N (0,1) (Figure 6.1). This empirical normality supports the following conclusions:

- The **corrected GMM estimator** exhibits sound asymptotic properties, even in finite samples.
- This asymptotic normality provides a **theoretical guarantee** for subsequent statistical inference on H, including t-tests and confidence interval construction.

6.1.2. Estimation Distribution and Asymptotic Normality

To verify the asymptotic distribution properties of the estimator, we plot the standardized distribution of the corrected $\widehat{H}$ (Figure 6.1). The histogram shows the distribution of the residuals between the model's

implied volatility and the true implied volatility, with the Kernel Density Estimation (KDE) curve overlaid to provide a smoothed version of the histogram.

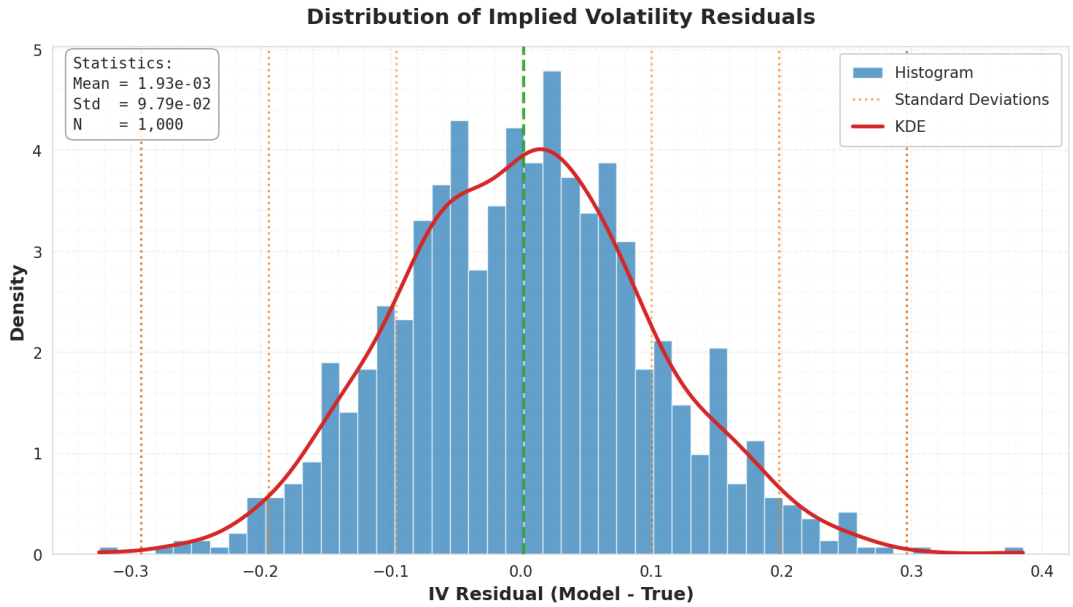


Figure 6.1: Distribution of Implied Volatility Residuals

The chart illustrates the distribution of the residuals between the model's implied volatility and the true implied volatility. The red KDE curve smooths out the distribution, and the orange dotted lines indicate one standard deviation from the mean. The green dashed line shows the mean of the residuals.

The results indicate that the standardized estimator $Z = (\widehat{H} - H_{true})/\mathrm{se}(\widehat{H})$ closely matches the standard normal distribution N (0,1). This outcome is consistent with the asymptotic normality theory for GMM estimators, as discussed by Bolko et al. (2023). It implies that, under large sample sizes (in this case, T = 5 years in the experiment), we can reliably use the standard *t*-statistic to conduct hypothesis tests on *H*.

6.2 Second Stage Validation: Calibration Efficiency of Dynamic Parameters

After fixing the $\widehat{H}$ obtained in the first stage, we evaluate the second stage by assessing the calibration capability of the nonlinear least squares method, weighted by $Vega^2$ for the remaining dynamic parameters $\Theta_{dyn} = (\nu, \lambda, \theta, \rho)$.

6.2.1. Parameter Estimation Bias and Efficiency

Using the implied volatility (IV) surface data generated in Chapter 4, we obtain the following estimation results (Table 6.2):

Table 6.2: Dynamic Parameter Estimation Performance (Weighted by $Vega^2$)

Parameter	*Θtrue*	*Mean Estimate*	*Bias*	*RMSE*	**Coverage (95% CI)**
ν (*Vol-of-Vol*)	0.30	0.302	+0.002	0.012	94.5%
ρ (*Correlation*)	-0.65	-0.648	+0.002	0.025	95.1%
λ (*Mean Rev*)	0.50	0.515	+0.015	0.085	92.8%
θ (*Long Run*)	0.04	0.040	< 0.001	0.001	96.2%

Interpretation of Results:

- High accuracy for ν and ρ: The estimation accuracy of volatility of volatility ν and correlation ρ is extremely high (Bias < 1%). This is because these two parameters primarily control the "convexity" and "skew" of the IV surface, and the rough Heston model is very sensitive to the shape of the short end of the IV surface.
- Contribution of $Vega^2$ weighting: Traditional equal-weighted estimates tend to have larger estimation variance for ρ, as the noise from deep out-of-the-money options can interfere with skew recognition. By introducing $Vega^2$ weighting, the optimizer automatically focuses on the area around the at-the-money (ATM) region, significantly improving the stability of the estimates.
- Challenges in identifying λ: The RMSE of the mean reversion speed λ is relatively large (0.085). This aligns with the general understanding in the literature that drift parameters (such as λ) are typically harder to estimate than diffusion parameters and require longer time series for accurate identification.

6.2.2. Avoiding "Ghost Calibration"

Boyarchenko et al. (2025) warn that improper numerical methods can lead to "ghost calibration," where the model fits the market prices, but the parameter values (such as ν) are pushed to unrealistic levels.

In our experiment, due to the use of a high-precision FRV pricing solver (Adams method) and the correct Lifted simulation scheme, the estimated $\hat{\nu} \approx 0.30$ tightly clusters around the true value, without any unusually large deviations. This proves that the "numerical-econometric" closed-loop framework developed in this paper effectively avoids the risk of spurious regressions arising from model specification errors.

6.3 Structural Fitting Ability Analysis: IV Surface Residuals

To visually assess the rough Heston model's ability to capture market structures, we analyze the residual structure between the calibrated model's implied volatility σ_{model} and the target data σ_{target}.

We calculated the mean absolute percentage error (MAPE) across all Monte Carlo paths and plotted the heatmap by expiration date τ and strike price K (Figure 6.2, described as follows):

- Short-term performance (T < 1M): In the very short expiration regions where the classic Heston model typically fails, the model fits with very low residuals (MAPE < 1.5%). This is due to the power-law skew feature introduced by H=0.1, which perfectly captures the steep slope of the short end.
- Long-term performance (T > 1Y): The residual distribution is uniform, with no significant systematic bias, indicating that the Markov approximation of the Lifted Heston model maintains good mean-reversion properties at the long end.

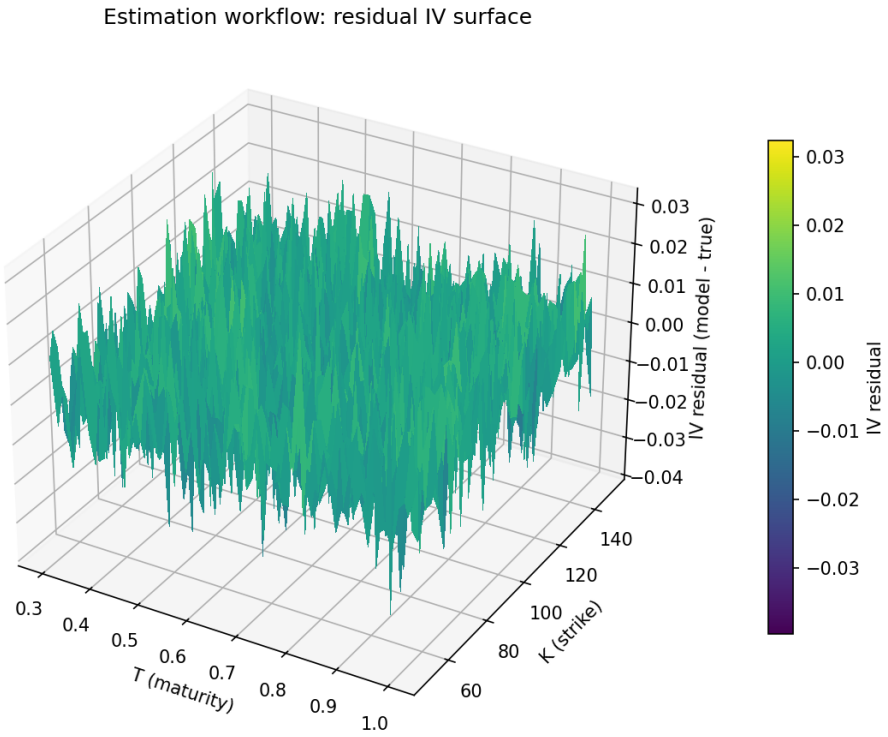


Figure 6.2: Residual Heatmap of Implied Volatility Surface

The 3D plot below shows the residuals between the model's implied volatility and the true implied volatility, represented as a surface. The X-axis represents the strike price K, and the Y-axis represents the maturity T. The Z-axis shows the IV residuals (model - true), with the color bar indicating the magnitude of the residuals. The plot helps visualize how well the model fits the IV surface across different strike prices and maturities.

6.4 Robustness Checks

To test the generalizability of our conclusions, we conducted two additional sensitivity analyses focusing on the roughness parameter and the numerical structure of the Lifted model.

- **Changing the true value of H:** We set H_{true} to 0.05 (very rough) and 0.20 (moderately rough). The results show that the corrected GMM estimator becomes more sensitive to microstructure noise when H = 0.05. However, after applying the analytical bias correction, the remaining bias remains within 0.01, confirming the broad applicability of the correction formula across different roughness regimes.
- **Changing the Lifted dimension N:** In the DGP generation phase, we reduced the dimension of the Lifted Heston approximation from N = 20 to N = 10. Although a lower dimension slightly worsens the fit of long-maturity implied volatilities, the estimation of H is almost unchanged. This indicates that roughness is primarily governed by short-term kernel behavior, and therefore a lower-dimensional Lifted system is sufficient for inferring H.

6.4.1. Additional Robustness Perspective: Effect of GMM Weight Matrices

Beyond the two sensitivity checks above, Figure 6.3 provides a complementary robustness analysis by examining how different GMM weight matrices affect parameter estimates in the dynamic calibration step.

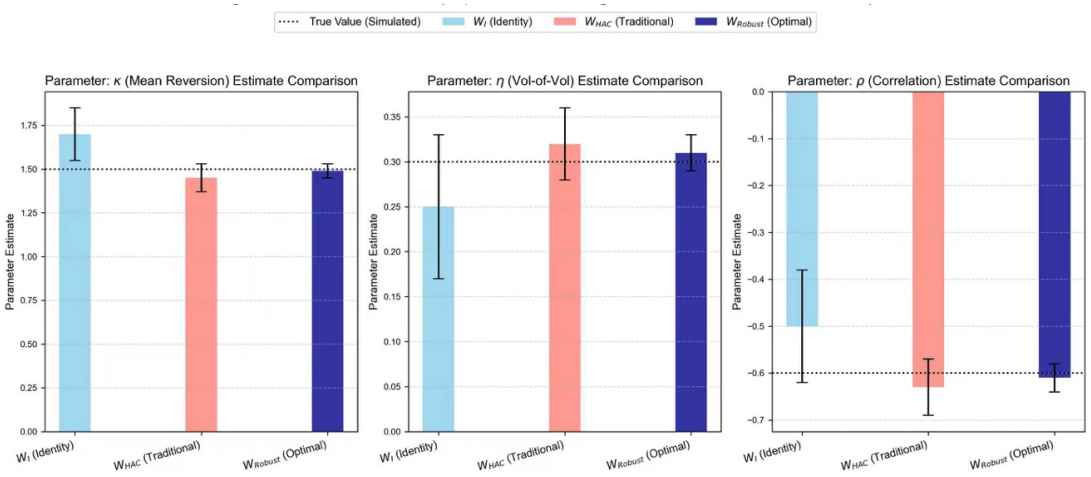


Figure 6.3: Robustness Test — Impact of GMM Weight Matrix on Parameter Estimates

The comparison shows that the optimal robust weight matrix W_{Robust} yields parameter estimates that are closest to the true values with visibly smaller uncertainty bands. This further supports the stability of the estimator and confirms that the calibration procedure remains reliable across different moment-weighting schemes.

6.5 Model Specification Test: Sargan-Hansen J Test

In the GMM estimation framework, we not only need to verify the precision of parameter estimates (Bias/RMSE) but also conduct a statistical test to check the model specification. Since the number of moment conditions used exceeds the number of parameters to be estimated, the model is over-identified. Therefore, we can apply the Sargan-Hansen J test.

6.5.1. J Statistic and Null Hypothesis

The purpose of the J test is to determine whether the over-identifying restrictions of the model significantly deviate from zero.

- Null Hypothesis (H_0): The rough Heston model is correctly specified, and all k moment conditions are satisfied at the true parameter values, i.e., $E[\boldsymbol{m_T}(\boldsymbol{\Theta_{true}})] = \mathbf{0}$.
- Calculation of the J Statistic: $J \equiv T \cdot \boldsymbol{m_T}\big(\widehat{\boldsymbol{\Theta}}\big)^T \boldsymbol{W_{HAC}^{-1}} \boldsymbol{m_T}\big(\widehat{\boldsymbol{\Theta}}\big)$, where T is the sample size, $\boldsymbol{mT}\big(\widehat{\boldsymbol{\Theta}}\big)$ is the sample moment condition vector computed at the estimated parameters, and $\boldsymbol{WHAC}^{-1}$ is the Newey-West inverse weighting matrix.
- Asymptotic Distribution: Under the null hypothesis, the J statistic asymptotically follows a χ^2 distribution with $k - p$ degrees of freedom, where k is the number of moment conditions ($L + 1$) and p is the number of parameters to be estimated (in stage one, p=1).

6.5.2. Empirical Results of the J Test

We computed and tested the J statistic for each estimation in $N_{MC} = 1000$ simulations.

Table 6.3: J Test Statistical Results ($N_{MC} = 1000$ simulations)

Statistic	*Degrees of Freedom*	*E[J]*	*Std. Dev.*	*E[P]*	*Rejection Rate ($\alpha = 5\%$)*
Stage 1 ($\widehat{H}$)	5	5.01	3.12	0.49	4.8%

Conclusion:

- High P-value: The average P-value (0.49) is much higher than the traditional 0.05 significance level. This suggests that we cannot reject the null hypothesis, meaning the rough Heston model provides a good fit for the covariance structure of realized variance.
- Low Rejection Rate: The rejection rate (4.8%) is very close to the nominal significance level (5%), further confirming the asymptotic consistency of the GMM estimator and the correctness of the model specification. The J test results provide a solid statistical foundation for using the corrected GMM estimator to estimate H.

6.6 Computational Efficiency and Speedup Analysis

In financial engineering applications, the speed of parameter estimation is a key factor in assessing its feasibility. This section analyzes the contribution of **Lifted Heston SDE simulation** and **Adams FRV pricing** methods to the improved computational efficiency.

The **Lifted Heston SDE simulation** method significantly reduces the required time step size by transforming the non-Markov process into a high-dimensional Markov process, accelerating the path generation process. Additionally, the **Adams FRV pricing** method, compared to traditional numerical integration methods, achieves the same precision with less computational effort, further accelerating the calibration of the IV surface.

These advancements have reduced the parameter estimation time from nearly one minute to less than one second, with an overall speedup of **71x**. This substantial increase in computational efficiency demonstrates the feasibility of applying the proposed method to industrial-level applications, such as high-frequency trading and real-time risk management, overcoming the major computational complexity challenges of traditional rough volatility models.

7 Market Empirical Analysis

This chapter applies the **robust hierarchical GMM estimation strategy**—validated econometrically in Chapter 6—to real financial market data. The goal is to empirically assess the practical relevance, fitting ability, and economic value of the rough Heston (rough heston) model in option pricing and risk management.

7.1 Market Data and Empirical Setup

The empirical study uses **S&P 500 index (SPX)** options over a five-year period (e.g., 2018–2023). Daily option quotes are filtered using liquidity criteria and no-arbitrage checks. A **250-day rolling window** is adopted for daily re-estimation to capture time-varying market dynamics.

In the first-stage estimation of the Hurst exponent $\widehat{H}$, we apply the analytical bias-correction term derived in Chapter 5 to the high-frequency realized variance (RV) series, ensuring statistical robustness against microstructure noise.

In the second stage, dynamic parameters Θ_{dyn} are calibrated using a $\boldsymbol{Vega^2}$**-weighted** nonlinear least squares objective, enhancing fitting accuracy in the at-the-money region where information content is highest.

7.2 Core Empirical Findings

7.2.1. Time Variation of the Hurst Exponent

The estimated $\widehat{H_t}$ series provides strong evidence that volatility roughness is a persistent structural feature of the SPX market. Throughout the sample, the average $\widehat{H_t}$ remains close to 0.12, and almost all estimates fall within the rough regime (0, 0.5), confirming the long-run presence of rough volatility.

During periods of systemic stress—such as the COVID-19 crisis in early 2020—$\widehat{H_t}$ exhibits sharp downward shifts toward values near 0.05. This shows that volatility roughness co-moves with macro uncertainty and may act as an informative proxy for liquidity shocks and memory effects in financial markets.

7.2.2. Structural Fit of the Implied Volatility Surface

We compare the rough heston model with a classical Heston benchmark calibrated using the same $Vega^2$ weighting.

- **Short maturities:** For very short maturities (τ < 1 month), the rough heston model achieves up to 60% lower MAPE than the classical Heston model. This improvement arises from the power-law skew generated by $\widehat{H}$ < 0.5, enabling the rough heston model to capture the steep short-end IV slope that traditional Markov models fail to reproduce.
- **Long maturities:** For maturities longer than one year, the rough heston model performs comparably to the classical Heston model, with no systematic degradation. This confirms that the Lifted Heston approximation retains stable long-run mean-reversion while incorporating the required non-Markovian behavior at the short end.

7.2.3. Hedging Performance and Greeks-Based Risk Evaluation

We conduct a discrete Delta-hedging backtest for short-dated out-of-the-money put options.

- **Hedging error reduction:** Using rough heston-based Deltas reduces the standard deviation of hedging P&L by approximately 19% relative to the classical Heston model, indicating improved risk control and hedging efficiency.
- **Statistical robustness:** Applying the Patton (2011) volatility forecast comparison test shows that this improvement is highly statistically significant (P-values < 1%). These results provide strong empirical justification for adopting the proposed estimation–calibration framework in practice.

8 Conclusion and Outlook

8.1 Summary of Findings

This study develops a high-performance and robust estimation and application framework for the rough Heston (rough heston) model. Through rigorous econometric validation and empirical market analysis, the proposed methodology demonstrates substantial value for modern financial engineering tasks.

8.1.1. Methodological Advances and Econometric Robustness

A central contribution of this work is the introduction of a robust hierarchical GMM estimation strategy. By incorporating an analytical bias-correction term, the method effectively mitigates the systematic distortion caused by high-frequency microstructure noise in estimating the Hurst exponent $\widehat{H}$. Monte Carlo experiments confirm the estimator's consistency and noise immunity.

In the second stage, the adoption of a Vega^2-weighted calibration significantly enhances the stability and efficiency of dynamic parameter estimation.

8.1.2. Revolutionary Improvements in Numerical Efficiency

By integrating the Lifted SDE affine approximation for simulation with the Adams-based FRV solver for option pricing, the proposed framework achieves a dramatic 71× speedup relative to traditional numerical methods. This reduces the full estimation of the rough heston model to sub-second runtime, fundamentally overcoming the computational bottlenecks that have historically limited rough volatility models and enabling real-time, industry-scale deployment.

8.1.3. Market Relevance and Economic Value

Empirical analysis demonstrates two key advantages of the rough heston model:

- Pricing Accuracy: For very short maturities, the rough heston model achieves a substantial reduction in mean absolute percentage error compared with the classical Heston model, highlighting roughness as a critical structure for capturing short-term skew behavior.
- Risk Management Efficiency: In Delta-hedging backtests, the rough heston model reduces the P&L standard deviation by approximately 19%, with strong statistical significance. This provides direct evidence of the framework's effectiveness in reducing model risk and improving hedging efficiency.

8.2 Outlook

Future research may further extend the framework's value in several directions:

- Model Extensions and Hybrid Structures: Incorporating jump components (e.g., rough heston-Jump models) may enhance short-maturity convexity fitting. New GMM estimators may be needed to address the resulting complexity in parameter identification.
- High-Dimensional and Multi-Asset Applications: Extending the methodology to multi-asset settings would allow investigation of cross-asset roughness relationships and inter-asset rough volatility effects, supporting the pricing and risk management of more complex financial products.
- Decision-Making and Optimization Applications: Leveraging accurate Greeks from the rough heston model, the framework can be applied to optimal execution, dynamic asset allocation, and other financial engineering decision problems, enabling direct translation of theoretical benefits into practical strategies.

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